

2802ICT Intelligent Systems

RUBRIC FOR ASSIGNMENT 2: NEURAL NETWORKS

Task: Classification Using Neural Networks (30 marks)

Part A — Coding (15 marks)

Criterion	Excellent	Good	Satisfactory	Poor
Code Correctness — Softmax Regression (784-10)	(3.5-4) Forward pass, loss, and gradient update correctly implemented; runs without errors; matches expected output format	(3-3.5) Mostly correct with minor bugs not affecting core functionality	(2-2.5) Partially correct; produces output but with logical errors	(0-1.5) Non-functional or missing
Code Correctness — Two-layer MLP (784-30-10)	(3.5-4) Sigmoid activation and hidden layer correctly implemented; forward/backward pass correct	(3-3.5) Minor implementation issues	(2-2.5) Partial implementation with notable errors	(0-1.5) Non-functional or missing
Code Correctness — Three-layer MLP with Residual Connection (784-30-30-10)	(4.5-5) Residual connection correctly implemented between hidden layers; activations and dimensions correctly handled	(3.5-4) Mostly correct with minor issues in residual connection logic	(2.5-3) Residual connection attempted but flawed or incomplete	(0-2) Not implemented or fundamentally incorrect
Use of Autograd & Command-line Flexibility	(2) Correct use of autograd; all specified command-line arguments function correctly	(1.5) Autograd used correctly; minor issues with argument flexibility	(1) Autograd usage mostly correct; several arguments hard-coded or non-functional	(0-0.5) Autograd misused or manual gradients implemented instead
Code Documentation (assessed alongside with coding)	Clear, meaningful comments throughout, especially at NotImplemented/Error sections; code is readable and well-structured	Adequate comments; mostly clear	Minimal or unclear comments	No meaningful documentation

Part B — Report (15 marks; including experiments)

Software Design Description (forward pass, loss computation, parameter updates)	(3) Clear, accurate, detailed explanation of how forward pass, loss, and updates work in the implementation	(2-2.5) Solid explanation with minor gaps	(1.5) Superficial or partially accurate description	(0-1) Missing or largely incorrect
Experiment (a): Baseline Network Training	(3) All four networks tested on all three datasets; results clearly	(2-2.5) Most combinations tested and	(1.5) Some combinations	(0-1) Experiment largely incomplete or

Criterion	Excellent	Good	Satisfactory	Poor
	tabulated/graphed; meaningful comparison across networks and dataset sizes	presented; minor gaps in comparison	missing; limited comparison	results not presented
Experiment (b): Learning Rate Analysis	(3) All five learning rates tested; correctly identifies best-performing and unstable rates with clear supporting evidence (graphs/tables)	(2-2.5) Most learning rates tested; reasonable identification of best/unstable rates	(1.5) Some learning rates missing or conclusions unsupported by data	(0-1) Experiment not conducted or conclusions absent/incorrect
Experiment (c): Mini-batch Size Study	(3) All four batch sizes tested; correctly identifies best accuracy and slowest training with clear supporting evidence	(2-2.5) Most batch sizes tested; reasonable conclusions	(1.5) Some batch sizes missing or conclusions unsupported	(0-1) Experiment not conducted or conclusions absent/incorrect
Analysis & Discussion	(2.5-3) Deep, well-reasoned explanation of observed patterns (network depth/residual effects, learning rate convergence behavior, batch size trade-offs); connects results to underlying theory	(2) Solid explanation with adequate reasoning; some depth missing	(1.5) Superficial or descriptive-only discussion lacking real analysis	(0-1) Little or no analytical discussion
Conclusion (assessed alongside with report)	Concise, accurate summary that ties together findings from all experiments	Adequate summary with minor gaps	Vague or incomplete summary	Missing or irrelevant
Screenshots & Reproducibility (assessed alongside with report)	Clear screenshots showing code running successfully for key experiments; output format matches expected style	Screenshots present but incomplete or unclear	Minimal/partial screenshots	Missing

Task total: 30 marks

Additional Considerations

Criterion	Notes
Individual/Authentic Work	Evidence of independent work; penalties apply for plagiarism per university policy
One-on-one Demonstration	Marks may be adjusted based on the student's ability to explain and demonstrate understanding of their own code and report during the conversation
Submission Compliance	Correct file naming (firstname_surname_Snumber_A2_report, firstname_surname_Snumber_A2_code.zip); correct file formats (.docx/.pdf for report, .zip for code)

Criterion	Notes
Restricted Library Use	Deduction if prohibited libraries, such as sklearn, pytorch, tensorflow, are used to implement core algorithms
